

Morgridge Institute launches platform to simplify FLIM data analysis

Researchers from the Melissa Skala Lab at the US-based Morgridge Institute for Research have developed FLIM Playground, a new open source platform designed to simplify and standardise fluorescence lifetime imaging microscopy (FLIM) data analysis. Published in *Cell Reports Methods*, the tool provides an end-to-end environment for processing, visualising and analysing complex cell imaging data.

FLIM is widely used to study cell metabolism, cancer treatment responses, autoimmune diseases and molecular dynamics. However, analysing FLIM datasets typically requires multiple software packages, custom scripts and extensive quality-control procedures. FLIM Playground addresses these challenges through a single graphical user interface that reduces the need to move data between different tools or repeatedly modify Python code.

Researchers can import FLIM images and cell masks, extract lifetime information, upload datasets, filter results, visualise patterns and perform advanced analyses such as dimensionality reduction and classification. Changes made within the interface are reflected instantly.

“In data analysis, especially FLIM data, there are so many settings you can adjust, but normally you have to go back into the Python code, change it, and re-run it to see the results,” said Wenxuan Zhao, lead developer and first author of the study. “It takes a lot of time and expertise. This lets you explore them on the fly—once you adjust something in the graphical user interface, the results are available instantly.”

The datasets undergo multi-source verification, temporal and geospatial correlation, equipment identification and attribution analysis before being incorporated into Swarmer’s AI training pipeline. Both companies noted that autonomous systems are only as reliable as the data used to train them, while unverified or manipulated data can introduce operational risks.

“This important partnership with Molfar Intelligence gives Swarmer a verified, battlefield-sourced data pipeline that no synthetic dataset can replicate. We believe that the quality of our AI models will improve considerably with Molfar’s methodology and that this will help with speed and accuracy of decision-making,” said Serhii Kupriienko, global CEO of Swarmer.

Swarmer’s autonomy software has supported more than 100,000 real-world combat missions in Ukraine since April 2024.

Chainguard-led Athena uses AI to secure open source software

A new industry coalition led by cybersecurity startup Chainguard has launched an AI-powered initiative aimed at finding and fixing vulnerabilities in open source software that underpins much of the world’s digital infrastructure.

Called Athena, the coalition brings together more than two dozen organisations, including JPMorgan Chase & Co., Cisco Systems Inc., Cloudflare Inc., and a range of cybersecurity firms. The group’s primary mission is to strengthen the security of open source software by identifying and remediating flaws discovered by advanced AI models.

The initiative focuses on software that serves as the foundation for critical technologies used across the economy, including web browsers, data centres, smartphones, ATM machines, and other essential digital systems.

Athena will leverage cutting-edge AI systems to uncover software vulnerabilities that may otherwise go undetected. Once identified, the flaws will be addressed through coordinated remediation efforts involving participating organisations and security experts.

The coalition reflects growing industry confidence in AI as a tool for improving software security and vulnerability management.

EXANTE’s €1M infrastructure fund will strengthen critical financial systems

Global prime broker EXANTE has launched Gecko Fund, a €1 million grant programme aimed at supporting critical open source software projects that underpin trading systems, financial data infrastructure, and broader financial technology ecosystems.



The initiative comes as concerns grow over the sustainability of open source software, which forms the backbone of many modern technology and financial services platforms. While businesses increasingly rely on open source technologies, maintainers of many critical projects continue to operate with limited financial support, creating potential risks related to software resilience, cybersecurity, and operational stability. The challenge is becoming more pressing as AI accelerates the discovery and exploitation of software vulnerabilities.

Gecko Fund will provide direct funding to maintainers of widely adopted but underfunded open source technologies, including APIs, developer tools, core infrastructure libraries, and software that enables modern financial services.

The programme will prioritise established projects that are critical to financial infrastructure yet receive limited funding relative to their importance. Focus areas